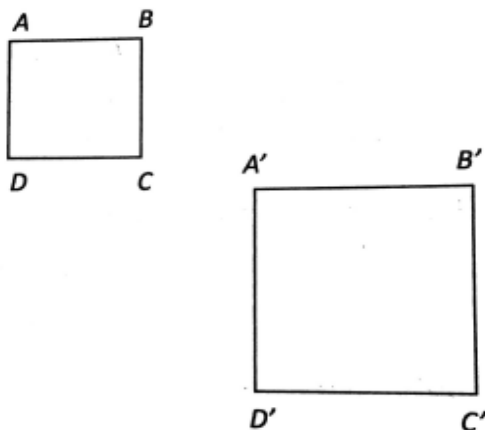


AMTI (NMTC) - 2012

BHASKARA CONTEST - FINAL - JUNIOR LEVEL

1. H is the orthocenter of an acute angled triangle ABC, with circumcentre O. Let P be a point on the arc BC not containing A of the circumcircle different from B and C. Let D be a point such that AD = PC and AD parallel to PC. Let K be the orthocentre of the triangle ACD. Prove that K lies on the circumcircle of triangle ABC.
2. Find all positive integer solution of the equation $4x^3 - 3x - 1 = 2y^2$.
3. Consider the set A of numbers $\left\{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{2012}\right\}$. We delete two of them say 'a' and 'b' and in their place we put only one number $a + b + ab$. After performing the operation 2011 times what is the number that is left over.
4. Seven digit numbers are formed by the digits 1, 2, 3, 4, 5, 6 and 7. In each number no digit is repeated. Prove that among all these numbers there is no number which is a multiple of another number.
5. ABCD and A'B'C'D' are two unequal squares in a plane placed as in the figure (A'B' parallel to AB etc.)



6. Find integers x, y, z such that $x^2z + y^2z + 4xy = 40$, $x^2 + y^2 + xyz = 20$.
7. There are two natural numbers whose product is 192. It is given that the quotient of the arithmetic mean to the harmonic mean of their greatest common measure and the least common multiple is $\frac{169}{48}$. Find the numbers.
8. Find all the positive integral solution of the equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{2013}$.
